Installation and Setup of CaDiCal

1 Official Source

The official source of the application is:

`https://fmv.jku.at/cadical/`

2 Institute Research Overview

The following image shows the research focus of the Institute for Formal Models and Verification, developers of CaDiCal.

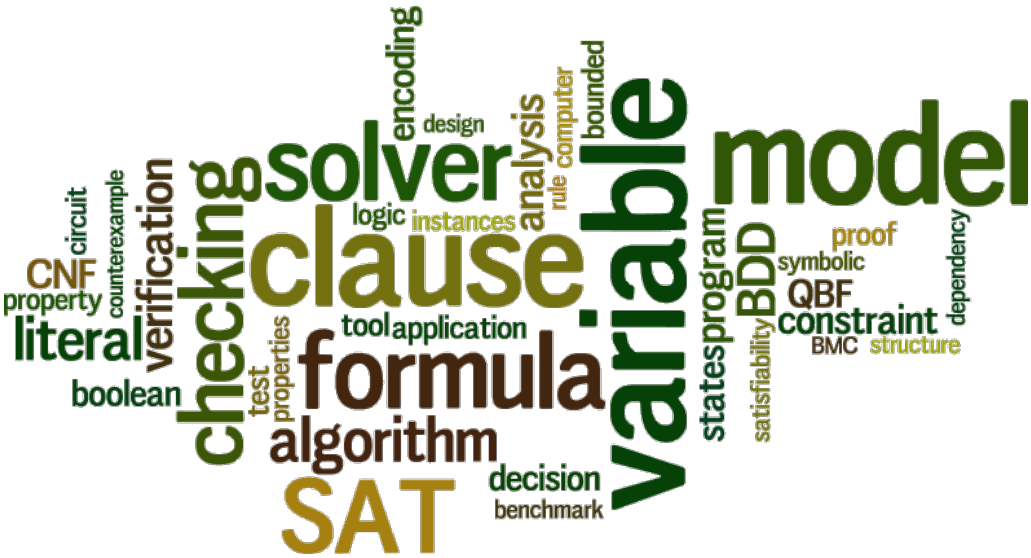


Figure 1: Research Focus Word Density

3 Required Tools

- C++ compiler (g++)
- make

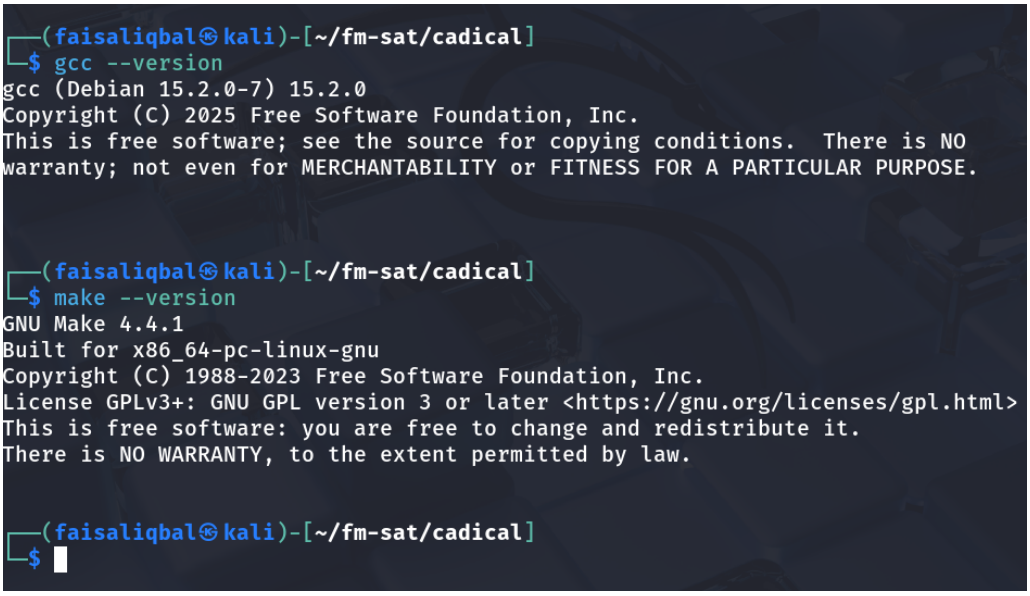


Figure 2: Verification of Prerequisites

4 Installation Steps

4.1 Cloning the Repository

```
git clone https://github.com/arminbiere/cadical.git
```

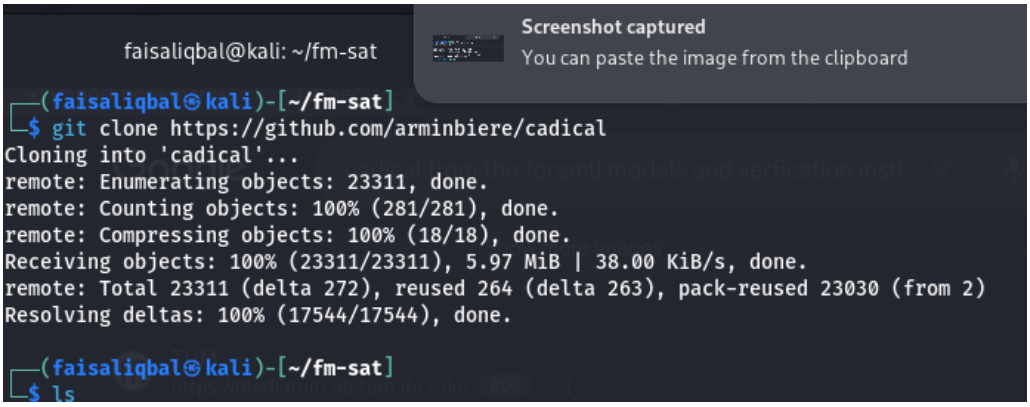


Figure 3: Cloning the Repository

4.2 Navigate to the CaDiCal Directory

```
cd cadical
```

4.3 Viewing Downloaded Files

After entering the directory, verify the downloaded files.

Figure 4: Directory of CaDiCal

4.4 Running the Configure Script

```
./configure
```

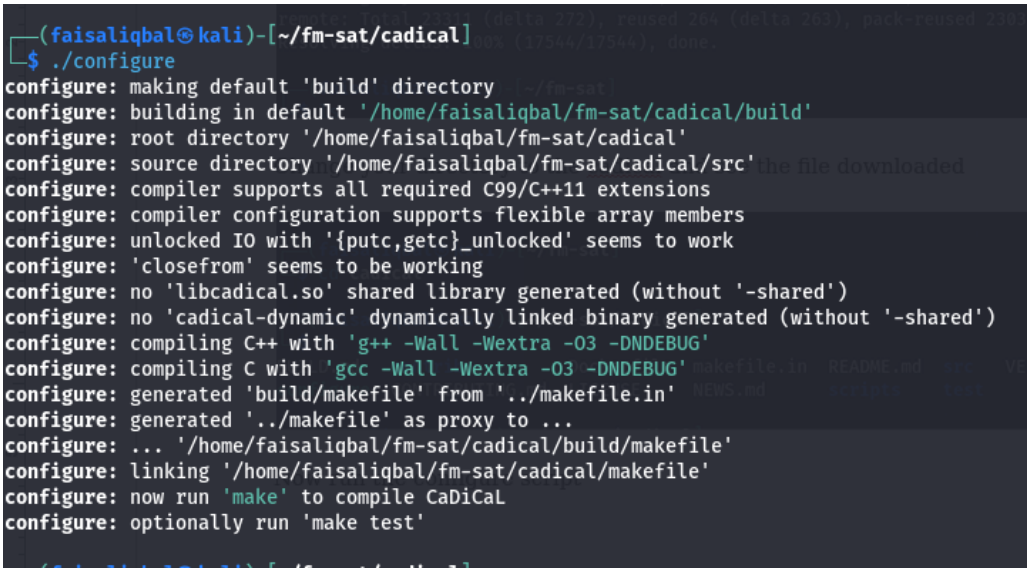
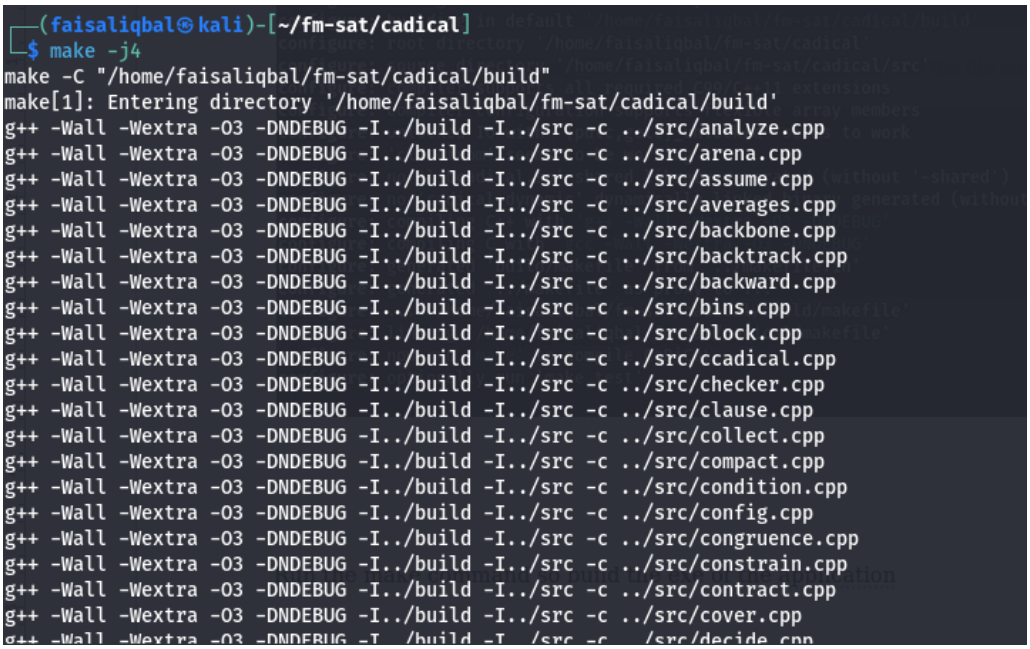


Figure 5: Running Configure Script

4.5 Building the Executable

make



```
┌──(faisaliqbal㉿kali)-[~/fm-sat/cadical]
└─$ make -j4
make -C "/home/faisaliqbal/fm-sat/cadical/build"
make[1]: Entering directory '/home/faisaliqbal/fm-sat/cadical/build'
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/analyze.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/arena.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/assume.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/averages.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/backbone.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/backtrack.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/backward.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/bins.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/block.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/ccadical.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/checker.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/clause.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/collect.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/compact.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/condition.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/config.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/congruence.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/constrain.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/contract.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/cover.cpp
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/decide.cpp
```

Figure 6: Building the Application with Make

4.6 Verifying Installation

```
./build/cadical --version
```

```
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -c ../src/version.cpp
ar rc libcadical.a analyze.o arena.o assume.o averages.o backbone.o backtrack.o backward.o bins.o block.o ccadical.o checker.o
 clause.o collect.o compact.o condition.o config.o congruence.o constrain.o contract.o cover.o decide.o decompose.o deduplicat
e.o definition.o drattracer.o elim.o elimfast.o ema.o extend.o external.o external_propagate.o factor.o file.o flags.o flip.o
format.o frattracer.o gates.o idruptracer.o instantiate.o internal.o ipasir.o lidruptracer.o limit.o logging.o lookahead.o lra
tchecker.o lrattracer.o lucky.o message.o minimize.o occs.o options.o parse.o phases.o probe.o profile.o proof.o propagate.o q
ueue.o random.o reap.o reduce.o rephase.o report.o resources.o restart.o restore.o score.o shrink.o signal.o solution.o solver
.o stable.o stats.o subsume.o sweep.o terminal.o ternary.o tier.o transred.o unstable.o util.o var.o veripbtracer.o version.o
vivify.o walk.o walk_full_occs.o warmup.o watch.o craigtracer.o kitten.o
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -o cadical cadical.o -L. libcadical.a
g++ -Wall -Wextra -O3 -DNDEBUG -I../build -I../src -o mobical mobical.o -L. libcadical.a
make[1]: Leaving directory '/home/faisaliqbal/fm-sat/cadical/build'

┌──(faisaliqbal㉿kali)-[~/fm-sat/cadical]
└─$ ./build/cadical --version
3.0.0

┌──(faisaliqbal㉿kali)-[~/fm-sat/cadical]
└─$
```

Figure 7: Verifying Installation

5 Running CaDiCal

After successful installation, create a `.cnf` file and run:

```
./build/cadical yourfile.cnf
```

6 Issues Encountered

Initially, the tool was tested on Windows. The build failed because:

- C++ was not installed.
- Required build system tools were missing.

Even after installing C++, the build still failed due to missing dependencies.

The environment was then switched to Linux. On Linux, installation required only three main commands:

- Clone the repository
- Run configure
- Run make